Lab Report
CSE 302 – Computer Organization & Architecture

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Task#1: Write a program to ask the user to enter two integers **A** and **B** and then display the result of computing the expression: **A + 2B - 5**.

```
.data
msgA: .asciiz "Enter A: "
msgB: .asciiz "Enter B: "
msgR: .asciiz "Result: "
.text
main:
li $v0,4
la $a0,msgA
syscall
li $v0,5
syscall
move $t0,$v0
li $v0,4
la $a0,msgB
syscall
li $v0,5
syscall
move $t1,$v0
sll $t2,$t1,1
add $t3,$t0,$t2
addi $t3,$t3,-5
li $v0,4
la $a0,msgR
syscall
li $v0,1
move $a0,$t3
```

```
syscall
```

Task#2: Assume that **`$s1 = 0x12345678`** and **`$s2 = 0xffff9a00`**. Determine the content of registers **`$s3`** to **`$s6`** after executing the following instructions:

```
.data
msgS3: .asciiz "\nS3: "
msgS4: .asciiz "\nS4: "
msgS5: .asciiz "\nS5: "
msgS6: .asciiz "\nS6: "
.text
li $s1,0x12345678
li $s2,0xffff9a00
and $s3,$s1,$s2
or $s4,$s1,$s2
xor $s5,$s1,$s2
nor $s6,$s1,$s2
li $v0,4
la $a0,msgS3
syscall
li $v0,34
move $a0,$s3
syscall
li $v0,4
la $a0,msgS4
syscall
li $v0,34
move $a0,$s4
syscall
li $v0,4
la $a0,msgS5
syscall
```

```
li $v0,34
move $a0,$s5
syscall
li $v0,4
la $a0,msgS6
syscall
li $v0,34
move $a0,$s6
syscall
```

Task#3: Assume that **`$s1 = 0x87654321`**. Determine the content of registers **`$s2`** to **`$s4`** after executing the following instructions:

```
.data
msgT2: .asciiz "\nS2: "
msgT3: .asciiz "\nS3: "
msgT4: .asciiz "\nS4: "
.text
li $s1,0x87654321
sll $s2,$s1,16
srl $s3,$s1,8
sra $s4,$s1,12
li $v0,4
la $a0,msgT2
syscall
li $v0,34
move $a0,$s2
syscall
li $v0,4
la $a0,msgT3
syscall
li $v0,34
move $a0,$s3
syscall
li $v0,4
la $a0,msgT4
syscall
li $v0,34
move $a0,$s4
syscall
```

Task#4: Write a program that asks the user to enter an alphabetic character (either lower or upper case) and change the case of the character from lower to upper and from upper to lower and display it.

```
.data
msgIn: .asciiz "\nEnter char: "
msgOut: .asciiz "Converted: "
.text
li $v0,4
la $a0,msgIn
syscall
li $v0,12
syscall
move $t0,$v0

blt $t0,97,upperToLower
bgt $t0,122,upperToLower
addi $t1,$t0,-32
j printCase
upperToLower:
blt $t0,65,printCase
bgt $t0,90,printCase
addi $t1,$t0,32
printCase:
li $v0,4
la $a0,msgOut
syscall
li $v0,11
move $a0,$t1
syscall
```

Task#5: Write a program that asks the user to enter and integer number and read it. Then ask him to enter a bit position (between 0 and 31) and display the value of that bit.

```
.data
msgNum: .asciiz "\nEnter number: "
msgBit: .asciiz "Enter bit position: "
msgVal: .asciiz "Bit value: "
.text
li $v0,4
la $a0,msgNum
syscall
li $v0,5
syscall
move $t0,$v0
li $v0,4
la $a0,msgBit
syscall
li $v0,5
syscall
move $t1,$v0
srlv $t2,$t0,$t1
andi $t2,$t2,1
li $v0,4
la $a0,msgVal
```

```
syscall
li $v0,1
move $a0,$t2
syscall
```

Task#6: Write a program that asks the user to enter a signed number and read it. Then display the content of multiplying this number by **24.5**.

```
.data
msgN: .asciiz "\nEnter signed number: "
msgM: .asciiz "Result: "
.text
li $v0,4
la $a0,msgN
syscall
li $v0,5
syscall
move $t0,$v0
sll $t1,$t0,4
sll $t2,$t0,3
add $t3,$t1,$t2
addi $t4,$t3, ($t0/2)
li $v0,4
la $a0,msgM
syscall
li $v0,1
move $a0,$t3
syscall
```

Task#7: Write a program that asks the user to enter an unsigned number and read it. Then swap the bits at odd positions with those at even positions and display the resulting number. For example, if the user enters the number **9**, which has binary representation of **1001**, then bit **0** is swapped with bit **1**, and bit **2** is swapped with bit **3**, resulting in the binary number **0110**. Thus, the program should display **6**.

```
.data
msgU: .asciiz "\nEnter unsigned: "
msgRes: .asciiz "Swapped: "
.text
li $v0,4
la $a0,msgU
syscall
li $v0,5
syscall
move $t0,$v0
li $t1,0xAAAAAAAA
li $t2,0x55555555
and $t3,$t0,$t1
and $t4,$t0,$t2
srl $t3,$t3,1
sll $t4,$t4,1
```

```
or $t5,$t3,$t4
li $v0,4
la $a0,msgRes
syscall
li $v0,1
move $a0,$t5
syscall
li $v0,10
syscall
```